QOSAIN BUKHARI

Junior Machine Learning Engineer

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SUMMARY

Software Engineering graduate (2026) with a 3-month ML internship and end-to-end projects in tabular ML, computer vision and RAG. I focus on honest evaluation (leakage checks, time-based splits, measured RAG metrics) and on shipping models as tested FastAPI services with Docker, MLflow, DVC and GitHub Actions. Seeking a junior ML engineer or ML internship role.

EDUCATION

Mirpur University of Science & Technology (MUST) **2022 – 2026**
BS Software Engineering | CGPA: 3.40/4.0

EXPERIENCE

Machine Learning Intern | Pixels Inn Institute, Mirpur **Jul 2025 – Oct 2025**

- Prepared datasets with Pandas and NumPy, fitting transforms on training data only to avoid leakage.
- Benchmarked linear, tree and ensemble models in Scikit-learn with cross-validation (F1, ROC-AUC, RMSE).
- Trained CNNs in TensorFlow and PyTorch and built an OpenCV + YOLO detection workflow; rated “Excellent” by the Managing Director.

PROJECTS

Cancellation Radar: Hotel Booking Cancellation Prediction **GitHub | 2026**
XGBoost, Optuna, SHAP, MLflow Registry, DVC, Pandera, Evidently, FastAPI, Docker

- Predicts booking cancellations at booking time (119,390 bookings). Removed 6 leaking check-in columns and tested on the newest 20% of arrivals: honest ROC-AUC 0.824.
- Set the decision threshold by business cost: catches 94% of cancellations and cuts the cost of wrong decisions by 76% (€40.86 → €9.69 per booking).
- FastAPI service with SHAP explanations and drift checks; MLflow promotes only models that beat production.

AI Accident Detection System (Final Year Project) **GitHub | 2025 – 2026**
YOLOv12, OpenCV, FastAPI, React, MySQL, Docker Compose, Pytest

- Trained a custom YOLOv12 model for 3 classes (fire, moderate, severe): 89.9% mAP@50, 92.5% precision and 86.8% recall on a 1,800-image held-out test set.
- Full system: FastAPI with JWT auth and MySQL, video jobs, email/WhatsApp alerts and a React dashboard.

Advanced Document Q&A RAG Chatbot **GitHub | 2026**
LangChain, ChromaDB, BM25, Cross-encoder, Groq LLM, Langfuse, FastAPI, Streamlit, Docker

- PDF chatbot with page citations, upgraded from naive RAG to parent-child chunking, hybrid BM25 + dense search (RRF), cross-encoder reranking, compression and query rewriting, each switchable in config.
- Built a 25-question evaluation (Recall@K, MRR, LLM-as-judge faithfulness): on the test PDF, naive retrieval scored best (Recall@1 0.96, faithfulness 0.95), so techniques stay configurable per corpus.
- 63 offline Pytest tests (fake LLM and embeddings) run on GitHub Actions CI.

Fake News Detection: MLOps Pipeline **GitHub | 2025**
Scikit-learn, NLTK, TF-IDF, MLflow, DVC, FastAPI, Docker

- DVC pipeline, MLflow tracking and a FastAPI endpoint; best of 4 classifiers on 72,134 articles: F1 95.8%.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Pandas, NumPy, Pandera, EDA, feature engineering
Machine Learning: Scikit-learn, XGBoost, Optuna, SHAP, PyTorch, TensorFlow/Keras, OpenCV, YOLO
LLM / RAG: LangChain, ChromaDB, hybrid search, reranking, RAG evaluation (LLM-as-judge), Groq, Langfuse
MLOps / Backend: FastAPI, Docker, MLflow, DVC, Evidently, GitHub Actions, Pytest, Git, MySQL

CERTIFICATIONS

Deep Learning Specialization, DeepLearning.AI (2025)
Generative AI, CampusX (2026)
Data Science, ML, DL & NLP Bootcamp, Udemy (2024)